

Aadith Thiruvangadan

Software Developer

Motivated Software Developer with skills in Python, SQL, and web development. Passionate about building efficient software solutions, designing systems, and continuously learning new technologies. Strong problem-solving abilities with a focus on delivering quality results.

aadiththiruvangadan@gmail.com +91-9744345247 Kannur, Kerala, India

About Me

I am a Computer Science and Engineering graduate (Class of 2025) with a strong foundation in modern software engineering practices. My expertise spans Python development, relational database queries, full-stack web applications, and artificial intelligence implementations.

I specialize in applying machine learning algorithms to practical problems, building clean user-centered web applications, and normalising database structures. I bring strong analytical thinking, self-reliance, and a commitment to writing clean, maintainable code to every engineering team.

Technical Skills

Programming Languages

Python, C, SQL

Core Concepts

Object-Oriented Programming (OOP), Data Structures, Algorithms

Web Development

HTML, CSS, JavaScript, PHP

Frameworks & Databases

Django, MySQL, Oracle SQL

Data Science & ML

Scikit-learn, XGBoost, Pandas, NumPy, OpenCV

Developer Tools

Git, GitHub, Power BI, Excel, VS Code

Projects

MACHINE LEARNING & REGRESSION ANALYSIS

YouTube Video Views Prediction using Machine Learning

An end-to-end Machine Learning pipeline utilizing text modeling and statistical attributes to forecast expected video view counts prior to publishing.

Problem Statement

Content creators lack tools to forecast how their video metadata (like titles) and initial engagement rates (like expected likes) will impact overall view performance. Predicting views beforehand helps content curators optimize titles and plan publication timings more effectively.

Solution Overview

Developed an analytical machine learning regression model. The system processes textual features of video titles using TF-IDF vectorization and merges them with numerical engagement estimators (likes). A tuned XGBoost regressor then predicts overall views, achieving strong prediction accuracy.

Individual Contribution

- Built the core Machine Learning model to predict video views using title and likes inputs.
- Performed data preprocessing, feature engineering, and robust text handling for title-based inputs.
- Applied TF-IDF text vectorization to transform title words into numerical feature weights.
- Implemented and optimized an XGBoost regression model, achieving a solid prediction performance rating (81% accuracy).

Tech Stack

Python, Scikit-learn, XGBoost, Pandas, NumPy

Measurable Impact

81% - Prediction Performance Accuracy **TF-IDF** - Advanced Title Text NLP Handling

XGBoost - Optimized Gradient Tree Boosting

Challenges & Learnings

Title Feature Engineering

Challenge: Raw video titles are highly arbitrary and contain non-standard punctuation and emojis, introducing noisy dimensions.

Solution: Deployed specific regex title cleaners, parsed emoji weights, and set max features limits on TF-IDF representation to filter out outlier text weights.

Like Count Covariance Bias

Challenge: Model relied too heavily on like numbers to guess view counts, resulting in erratic predictions for new videos with sparse initial likes.

Solution: Introduced L1/L2 regularization criteria inside XGBoost, forcing the model to weight text elements and likes more evenly.

Optimizing Regressor Performance

Challenge: Achieving target correlation thresholds under default hyperparameter constraints.

Solution: Conducted grid searches over learning rates, sub-sample ratios, and depth layers, raising test accuracy scores to 81%.

Education & Training

Academic & Professional Timeline

JUN 2019 – MAR 2021

Higher Secondary Education (HSC)

Govt HSS Kannadiparamba

Focus on Mathematics, Physics, Chemistry, and Computer Science fundamentals.

NOV 2021 – APR 2025

B.Tech in Computer Science Engineering

College Of Engineering Trikaripur

Specialized in Core Computer Science studies, including Data Structures, Algorithms, Object-Oriented Programming (OOP), Database Systems, and Machine Learning operations.

JULY 2025 – MARCH 2026

Python Fullstack and Data Science Training

Pyspiders / Qspiders

Intensive career training covering Core Python, Object-Oriented Programming (OOP), Django, REST APIs, HTML, CSS, JavaScript, Power BI, SQL, NumPy, Pandas, Statistics, and Machine Learning concepts.

Built foundational knowledge in full-stack development and data analysis.

 Certificates

NPTEL Certificate

Joy of Computing Using Python

Issued by **IIT Madras** upon completion of Python algorithm coursework and examinations.

PySpiders / Qspiders

Python Full Stack and Data Science

Comprehensive course certification covering full-stack web applications and machine learning analytics.

GitHub Project Repositories

youtube_views_predictions

Machine learning regression pipeline utilizing NLP TF-IDF text features and XGBoost to predict YouTube video view counts based on titles and likes.

detecting-humans-in-search-and-rescue-operations-

Real-time human detection system using YOLO and OpenCV, with a Django admin panel dashboard, MySQL log database, and Android client for push notifications.

Payroll-Management-System

A web-based administration application built to automate organizational salary distribution and tax calculations while providing secure self-service modules.

Contact & Conclusion

Let's Collaborate

I am currently seeking full-time opportunities in ****Software Development, Web Development, or AI-ML/Data Science Engineering****. I am ready to bring my dedication to code quality, logical thinking, and proactive problem-solving to a development team.

Feel free to contact me via email, explore my GitHub repositories, or connect on LinkedIn to discuss open developer roles, technical questions, or potential collaborations.

EMAIL ADDRESS

aadiththiruvangadan@gmail.com

PHONE NUMBER

+91-9744345247

LOCATION

Kannur, Kerala, India

LINKEDIN CONNECTION

linkedin.com/in/aadiththiruvangadan

GITHUB WORKSPACE

github.com/aadiththiruvangadan